

Module Code: CS3VR

Assignment report Title: VR Group World

Date (when the work completed): 17 Dec 2024

Actual hrs spent for the assignment: 30 hours

Group number: 2

Student numbers:

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Student name	What work has been completed	Level of contribution? (Fair, Limited or None)	Justification (if not Fair)
Derrick Chipepa	Created bedroom floor assets and associated logic. Created own assets for bedroom.	Fair	N/A
Jayden Scarpa	Created VR and PC cross platform interaction system for all interactables. Sourced prefab assets, materials. Created elevator and all associated floor-to-floor logic	Fair	N/A
Max Griggs	Created server floor and all associated logic. Created themed sound track	Fair	N/A
Mihai Visan	Created lobby and all associated logic. Created bedroom floor assets. Sourced feedback V/O audio. Created own assets for lobby.	Fair	N/A
Saif Naseem	Compiled report. Created some bedroom logic	Fair	N/A
Zahra Hashem	Created office floor and all associated logic. Sourced audio	Fair	N/A

Title and Aims of World

The title of the world is CyberQuest: Progression Zones. It is an immersive and educational virtual reality experience which is designed to engage learners with regard to basic computing logic and cybersecurity concepts. The primary aim of the virtual world is to create an engaging and gamified experience through the use of themed rooms alongside various computer-science-related concepts.

Four concepts/rooms were chosen, being the office, bedroom, server room and lobby:

Floor 1 is an office with interactive questions focused around basic coding keywords in pseudocode, helping to understand the key concepts in variable assignment and logic. Player interaction is achieved by solving multiple choice problems for fill in the blank questions. There is also the opportunity to explore the “dilapidated” office floor with associated sound, visuals and an overlooking balcony into the cyberpunk-styled surroundings.

Floor 2 is the bedroom, which is an escape room based on social security. This zone immerses players in a scenario with relation to social engineering and security flaws, as users have to identify clues/flaws from the items in the room in order to proceed. Not all items in the room are security flaws, which adds an element of critical thinking. Items that are flaws are identified by clicking/selecting them; flaws turn red, while secure items turn green.

Floor 3 is the server room, which presents users with a high-stakes cybersecurity simulation featuring multiple servers, each representing a unique system defence scenario. The user has to carefully analyse and select the correct security response from a variety of potential strategies. The room has three rows of servers, with the user having to view four questions and then select the correct answer out of the three options concurrently. If the user is incorrect, they must return to the elevator and respawn. The server room transforms abstract security concepts into a compelling, interactive learning experience that tests both knowledge and strategic thinking, by creating a high-pressure, consequence-driven environment.

The lobby is a room in which the user spawns in at the beginning of the experience. There are three panels that explain what the user has to do in each of the aforementioned rooms. There is a main monitor in the lobby which narrates that instructions for play, including other senses in the experience.

Background research

Educational technology highlights the potential of immersive learning experiences, particularly in computing and cybersecurity. Research by Lineas-Vargas and Cieza-Mostacero (2024) reveals that emotions play a significant role in shaping how individuals respond to various contexts. VR environments, by simulating real-life scenarios, can evoke these emotional responses effectively. For instance, creating an immersive setting like a high-pressure server room can elicit specific emotions from participants, enhancing both their engagement and their learning experience. The study goes on to state that VR is able to create a strong sense of presence, which makes the user feel as if they are actually in the virtual environment. CyberQuest takes this onboard by having themed rooms with audio and

multiple interactions in a rich skybox, making them emotionally present in the learning environments. This strong sense of presence can improve their focus and engagement with the security and logic content, which could be useful for a computer science student learning basic concepts. In another study done by Hamilton et al. (2020), it was found that VR can significantly improve learning efficiency compared to traditional methods of learning. By using VR, CyberQuest can help players understand computing logic and cybersecurity concepts in a more efficient manner. The study goes on to state that VR can increase learner engagement and motivation due to its interactive and immersive nature. Gamified elements and themed rooms in this project embody this by keeping players engaged and motivated to learn.

In terms of technical research, several XR interaction toolkits were evaluated, including Unity's built-in toolkit Unity (2023) and Meta's Interaction SDK (2024). Unity's toolkit integrates seamlessly with the new input system, supporting controller interactions via 'raycasts' or direct object manipulation in 3D space. In contrast, the Meta SDK is limited to Meta-based headsets, which could hinder compatibility with other devices in the future. As a result, Unity's built-in toolkit was chosen, utilising 'raycast interactables', albeit other types exist. 'Raycasts' were preferred due to their intuitive design, enabling players to interact from greater distances within a large virtual environment, using two 'wands' that adapt with distance, thereby enhancing accessibility.

In terms of textures and components, most textures were sourced from a central pack online to ensure consistency across levels, when working in a team. Also, some elements were imported as prefabs into each room for consistency and ease in implementation of complex logic to proceed through questions across each room/floor. Although, some prefabs were created to add originality and creativity in designing certain elements. Elements such as the safe, drawers, quiz boards/displays, glass windows, lighting sources and theme soundtrack were made to suit the world. See the References section for all object/skybox textures, prefab and audio packs used and imported.

Screenshots



Fig.1: Game lobby - explains the floors of game



Fig.2: Bedroom - player must find security flaws



Fig.3: Office - player must complete logic questions in code

Instructions on how to use world

The world is designed in a cross-platform architecture, with a version existing that runs on a VR headset, in addition to a PC. The headset used for development and testing was a Meta Quest 3, with the “MetaBuild.apk” file installed upon the headset. Cross-platform interaction was achieved using Unity’s new input system with a single source-of-truth for any interaction or movement and two input maps per device linking to this, allowing for better interoperability and future maintenance.

In terms of navigation, the user is able to walk around the room via the joystick on a VR headset, using a tunnelling vignette (‘blindners’) applied upon movement to reduce motion-induced sickness. On a PC, a player can use the WASD keys to move, along with the mouse to look around. Any camera movement is frame-rate independent to further reduce jarring movement and associated motion sickness. The sensitivity was set low such that the user has time to explore and make fine adjustments to their movement to interact with complex objects in each room.

For interaction on a PC, players use a central red reticle to interact with items by aligning it with the desired object using the mouse and clicking. This functionality is powered by ‘raycasts’, which are invisible lines projected from the player’s viewpoint or controller to detect objects in the environment. In VR, the same principle applies: controller ‘raycasts’ are used to project beams toward objects elastically, enabling players to hover over and select them with the controller trigger. Hover enter and exit methods, triggered by the rays from each controller, are implemented on ‘interactables’ to provide affordances such as visual feedback (turning the ray blue) and haptics upon moving the controllers around the world.

Reflection

Interactivity

This experience is designed to be highly interactive, which draws the user into a dynamic and immersive learning environment. Users are not passive observers, but active participants, as the virtual world is to be directly engaged with. By manipulating objects, solving puzzles, and making critical decisions, learners actively construct their understanding of computing and cybersecurity concepts. In addition, objects not directly related to level completion are interactive too, like drawers, and sound effects while moving around and exploring, encouraging exploration through each floor.

User Experience

This entire experience uses the power of virtual reality to create an immersive and engaging user experience that is easy to control and be part of. The user is transported into a futuristic world using consistently realistic visuals in the immediate world and skybox, and spatial audio. Ease of interaction is achieved using a frame rate independent reticle with smooth and low sensitivity movement. In VR, controllers provide haptics and coloured visuals when moving around the world hinting at what can/cannot be interacted with (affordances). This all comes together to create a consistent sense of presence, making learners feel as if they are truly part of the virtual world.

Design and Purpose

This is a virtual reality experience which is designed to immerse learners in a gamified environment which teaches the fundamental computing and cybersecurity concepts to the user with its distinctly themed rooms. All these rooms leverage the power of VR to create a captivating and interactive learning environment.

Key design elements embodying design and purpose respectfully include:

- *Immersion*
 - 3D environments which are consistently detailed with textures used from a central source, which create a sense of presence of engagement.
 - Spatial audio, which include audio cues on interaction, that enhance the overall experience and guide the user through the VR world. Even background (city) audio was spatialised in each floor to help make the environment more immersive.
 - Realistic interactions which allow the user to explore the environment and solve its challenges in a natural way with affordances and text guiding the user.
 - A spatially consistent, “depth-less” skybox with appropriate colours and visuals that fit in with the cyberpunk theme.
- *Gamified Learning*
 - Immediate feedback gives real-time feedback on user actions through visuals (and auditory feedback - e.g. on level completion), which reinforces learning and provides guided exploration.
 - The game has levels which gives the user a sense of purpose and accomplishment after they have successfully completed one level.
 - There is a strong sense of ‘purpose’ in the game created through the structure of topic-themed floors with clearly set out instructions both in the lobby and before the user enters a level/room; this reduces confusion during play and explains clearly what must be done.

The aim is to bridge the gap between theoretical knowledge and practical application. By immersing learners in realistic, futuristic scenarios, CyberQuest fosters a deep understanding of computing, cybersecurity, and social engineering principles. The interactive nature of the game encourages the user to problem solve and critically think about which action to take. All in all, this VR experience empowers the learner to become a more informed user through gamification of learning.

Reflection on What was Done and Future Improvements

The development of CyberQuest was a collaborative effort, which combines mostly imported assets and textures with some custom-built components created using ProBuilder in Unity. This was done so that focus was placed upon the interaction logic and intricacy of the levels in the game, as well as running it on VR. Work was divided with each person designing a floor, and others assisting and helping one another where their expertise and strengths lie. However, communication and task deadlines For future improvement, more original assets could be designed and implemented to achieve more consistently styled rooms that fit with the cyberpunk theme across levels, and provide better affordances for interaction. Additionally, levels of learning could be considered to alter questions in each room/topic to be more varied and user-configurable. Furthermore, for accessibility, implementing user-adjustable movement controls, such as mouse sensitivity and turn speed/blinder toggles/teleportation movement in VR, could further personalise the experience.

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